

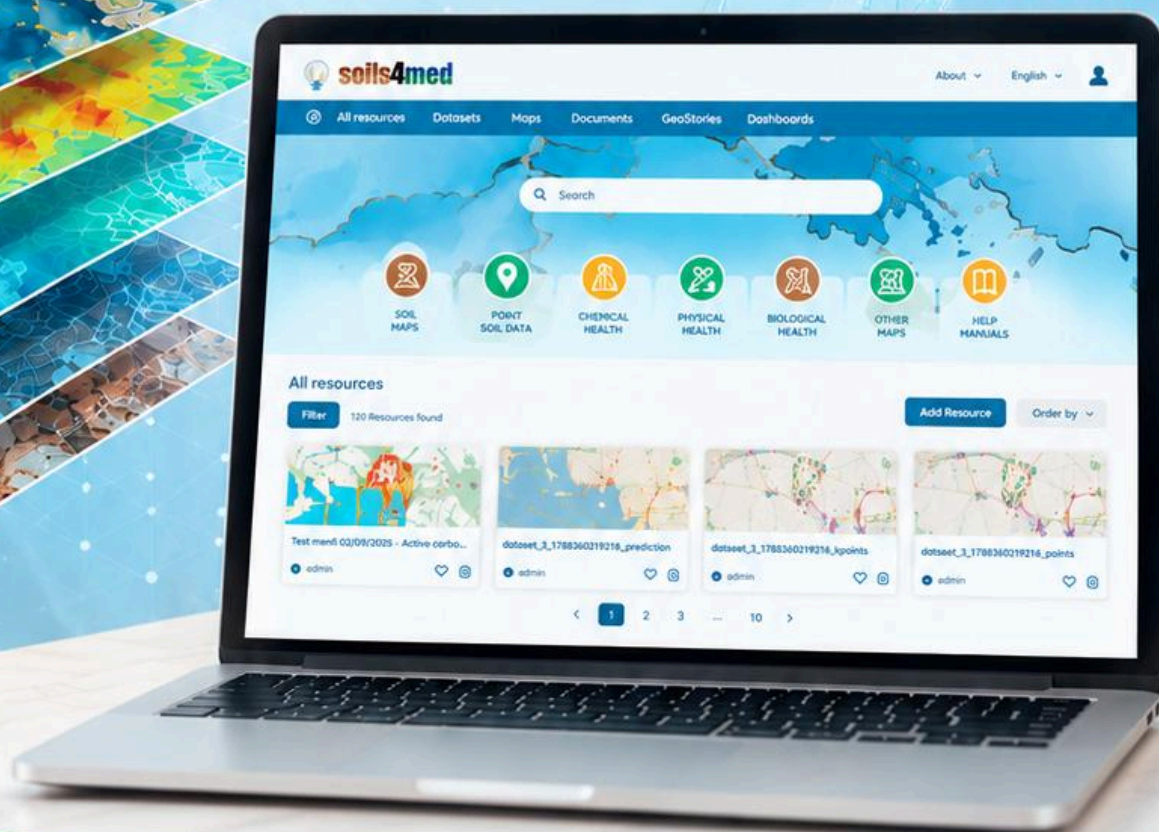
soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Instructions for the compilation of the point soil data .xlsx template

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The Soil Information System (SIS) that SOILS4MED implements needs to have the capacity to collect all the types of point soil data including legacy data gathered over time by the partner countries according diverse methods and formats, and at the same time provide a sound basis for data harmonization, interoperability, and intuitive consultation. For this reason we decided to adopt the field guide proposed by WRB 4TH EDITION as the international state-of-the-art reference, and we used the excel template sheets provided by the WRB as reference to create the project data model.

To maximize the advantages of the use of this reference, and to minimize the information losses implied by the conversion from the original data models to the new one, and to enable the upload of all the information generated by the project in the field, we introduced some adaptations.

In this document we explain in detail the data model developed by SOILS4MED by pointing out all the proposed changes. The latter mostly concern the insertion of new fields, the modification of some pre-existing ones and small changes to the attributes associated with some fields.

The changes implemented are reflected in the WRB soil profile description sheet attached to this document (S4M_soil_data_template_v2).

It is important to note that, for completing the S4M_soil_data_template, the field guide WRB 4TH EDITION (Annex 1) should be followed for the majority of fields, while this document should be used as a reference for the additional or modified fields.

- In addition to the two sheets originally included in the excel template provided by the WRB (i.e., “General and surface” and “Layer descriptions”) two other sheets (“Lab data” and “Lab data by sampling depth”), have been added. These, excepting the first three columns, are identical and should be used respectively to enter laboratory data related to samples obtained from the genetic horizons or from depth layers.
- In the “General and surface” and “Layer descriptions” sheets several fields were added or modified: newly inserted fields are shown with a blue background, while modified fields are shown with a light blue background and have the modified portions of the title, if any, written in red.
- In the case of new fields, all the information useful for compilation is provided by this document. If such fields are already described in other guides, the description and information provided by those guides taken as reference is also reported.
- For each dataset, the Excel file relating to the genealogy of the data must also be complete (S4M_data_genealogy_template_v2). The “Data genealogy” spreadsheet consists of a single sheet containing the essential metadata for the correct use of soil data.

Note: the chapters referred to are the WRB 4TH EDITION reference field guide chapters.

- New field: Point ID

Enter an ID for the point where soil information is gathered.

It will be composed of upper case letters and numbers (001, 002). It will not include spaces or special characters (only A-Z letters are accepted, along with numbers).

The ID must be different for each point data.

The ID must be as short as possible and respect different syntax rules, depending on the type of soil data:

For soil “legacy” data points, or any other point that is not supposed to be hit again in future in the frame of a regular monitoring process:

-Start with the ISO3166 code for subnational level – e.g. “Regione” in Italy, “Governorate” in Tunisia.

-Continues with “S4M”, to state that they were made available by SOILS4MED and are part of a coherent project datasets.

-Ends with a specific key ID assigned by Partner, made of letters followed by an incremental number.

Example: ESAN_S4M_RBN0001

Where ESAN means Andalucia

The specific key ID could coincide with the original ID (if any), that is the ID originally assigned to such point in the frame of a previous project. However, to ensure that all SIS ID are different, with no duplications, and that the syntax rules are respected, renaming can be necessary.

For soil monitoring points, that are supposed to be revisited in time for soil health-related monitoring:

-Start with the ISO3166 code as for legacy points

-Continues with “S4M” as for legacy points

-Ends with “SH” (Soil Health) followed by an incremental number and by the Year

Example: IT88_S4M_SH0001_2025

Where IT88 means Sardinia

- New field: Project ID

Enter a code that identifies univocally the project, or survey activity/campaign, in the frame of which the point data were generated. Please note that summary information about the original Projects must be entered in a dedicated project “Genealogy” file.

The Project ID must be assigned, even if the original project did not have one (e.g., no project acronym). The project ID must be as short as possible and respect the following syntax rules:

It will be composed of upper case letters and numbers (001, 002). It will not include spaces or special characters (only A-Z letters are accepted, along with numbers).

- New field: Original ID

Enter the original ID of the point data (typically, a soil profile or site ID). This is important to track back the lee

- Modified field: GPS coordinates in WGS84 SYSTEM (DEC.DEG.)

“in WGS 84 system” has been added to the field description.

NB: All coordinates must be entered in the WGS84 reference system and must be in decimal degrees.

The following rule should be used to convert a DMS coordinate to decimal degrees:

DMS = dd°mm'ss.sss"N DECIMAL DEGREE = dd + mm/60 + ss.sss/3600

DMS = dd°mm'ss.sss"S DECIMAL DEGREE = -dd + mm/60 + ss.sss/3600

DMS = dd°mm'ss.sss"E DECIMAL DEGREE = dd + mm/60 + ss.sss/3600

DMS = dd°mm'ss.sss"W DECIMAL DEGREE = -dd + mm/60 + ss.sss/3600

- New field: GPS

Define whether the point location was taken with a GPS tracker, or not.

Please note that information about location accuracy must be entered into the project “Genealogy” file mentioned above. This information is very important if a GPS was not used and more in general in all cases in which the location accuracy is uncertain.

See the table 1.3 to select the correct code.

Table 1.3. Information about GPS data.

Profile position taken using a gps tracker	Yes
Profile position acquired from a map	No
Unknown	Un

- Modified field: Elevation a.s.l.

“Elevation” replaced the term “Altitude” due to semantic accuracy.

The field description is unchanged

- New field: Elevation from DEM a.s.l.

Insert the elevation based on digital elevation model (DEM), if available.

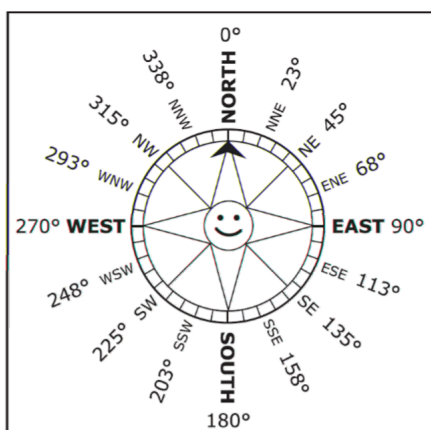
- Info for the compilation: Gradient upslope & gradient downslope

If the legacy gradient value available is related to the entire slope, enter that value in both fields.

- Info for the compilation: Slope aspect

The aspect value must be entered in degrees; in case of legacy data with cardinal points translate into degrees according to the Figure 1. 8.7 of field guide WRB 4TH EDITION (Annex 1), inserted below.

Figure 1,1. Slope aspect, Figure 8.7 of WRB 4TH EDITION (Annex 1) (Schoeneberger et al. (2012), 1-5)



Non-WRB Descriptor: Geomorphic features

In this new sub-chapter the landform affecting the area of the profile and which may have influenced its formation and evolution must be indicated.

Two types of landforms (“Landform 1” and “Landform 2”) and their degree of activity (“Activity of landform 1” and “Activity of landform 2”) can be reported. Moreover, a general description of the landscape can be inserted in the field “Description of geomorphic features”, if the data is available.

Below the instructions for filling in the fields.

- New fields: Landform 1 and Landform 2

Select among the landforms in table 1.4 those affecting the area of interest and enter the corresponding code. Use the help of a geomorphological map.

If the landform type is unclear, or you do not find a match with the proposed landforms, you can just use the general category (e.g. 'A' for Anthropogenic landforms; 'K' for Karst landforms; etc.) or sub-category (e.g. 'AA' for Area of accumulation).

IMP. If you only have a description of the morphology, or if you do not find a match with the proposed landforms, you can just fill in the free text field “Description of geomorphic features” (see field description below).

Table 1.4.

AA	Area of accumulation	AAW	Landfill of anthropic origin waste
		AAR	Landfill of remoulded natural material
AM	Mining area	AMQ	Abandoned quarry
		AMA	Active quarry
AC	Embankment or other constructions for water stream		

AT	Terraced slope	ATD	Deteriorated terraced slope
		ATI	Intact terraced slope
		ATM	Mechanizable terraced slope, with connected levels
AR	Grading, reshaped slope		
AS	Anthropic sinkhole		
KD	Karst depression	KDD	Sinkhole (Doline)
		KDU	Uvala
		KDP	Polje
		KDR	Residual relief (hum)
KK	Karren (field), grooves		
KG	Griza		
KF	Fluvio-karst valley	KFB	Blind valley
		KFC	Pocket valley
		KFD	Dry valley
KC	Cave		

SG	Gelifluction lobes		
SO	Soil slips deposit		
WH	Hanging valley step		
WV	V shaped small Valley		
WW	Wide fluvial valley		
WF	Flat bottom fluvial valley		
WG	Gorge		
WT	Fluvial terrace	WTE	Erosion scarp
		WTS	Surface
WL	Rills and gullies		
WB	Badlands	WBC	Calanco
		WBB	Biancane
WP	Earth pyramids		
WC	Glacis		
WA	Alluvial fan		
WD	Colluvial deposit		
WU	Alluvial plain	WUB	Reclaimed alluvial plain
		WUD	Depressed area in an alluvial plain
WR	Riverbed	WRB	Braided
		WRA	Anastomosing
		WRM	Meandering
		WRS	Single channel
		WRE	Ephemeral
		WRC	Crevasse splay
		WRR	River bank
		WRI	Fluvial island
		WRF	Floodplain and overflow area

WS	Spring depression		
GB	Block stream/field		
GR	Rock glacier		
GC	Glacier cirque		
GU	Crioturbation unevenness		
GH	Hummock		
GP	Poligonal soil		
GF	Fluvio-glacial deposit	GFE	Esker
		GFK	Kame terrace
		GFS	Sandur
GV	Glacial valley	GVH	Hanging glacial valley
GD	Glacial deposit	GDS	Subglacial deposit
		GDA	Ablation moraine
		GDF	Lateral/Frontal moraine
		GDI	Intermoraine depression
		GDD	Drumlin
GN	Nivation niche		
GS	Bowl shaped depression	GSI	Infilled bowl shaped depression
GT	Glacial terrace		
GA	Snow Avalanche channel	GAC	channel
		GAD	deposit
GO	Protalus ramparts		
MA	Abrasion platform	MAC	Cliff foot (talus)
ML	Lacustrine terrace		

MP	Lacustrine plain	MPM	Mineral deposit infill
		MPO	Organic deposit infill (peat bog)
MR	Reclaimed draining lacustrine plain	MRM	Mineral deposit infill
		MRO	Organic deposit infill (peat bog)
MC	Coastal plain	MCT	Tidal plain
		MCR	Coastal ribbon
		MCM	Mud plain
		MCC	Tidal canal
		MCS	Sand plain
MT	Marine terrace		
MO	Lagoon		
MD	River delta	MDR	Reclaimed delta
TC	Cuesta		
TD	Structural depression (graben)		
TG	Irregular slopes due to tectonic structure influence		
TR	Structural relief (horst)		
TS	Structural surface	TSD	Dissected structural surface
		TSW	Wavy structural surface
TV	Fault slope		
VA	Caldera		
VR	Volcanic crater	VRM	Maar crater
VC	Volcanic cone	VCA	Volcanic ash cone
		VCL	Volcanic lava cone
		VCP	Polygenic volcanic cone
		VCS	Scoria volcanic cone
VD	Lava dome		
VL	Lava flow		
VH	Lahar deposit		
VF	Pyroclastic flow deposit		
VP	Volcanic Plateau		
VS	Structural – Volcanic depression		

ES	Aeolian deflation surface		
ED	Aeolian deposit	EDL	Loess
		EDE	Dune affected by erosion
		EDD	Dune
		EDS	Stabilized dune
		EDA	Dune affected by anthropic influence
EI	Interdune area	EIL	Interdune area periodically inundated

- New fields: Activity of Landform 1 and Activity of Landform 2

If available, insert the code corresponding to the degree of activity of the processes that shaped the landscape (table 1.5).

Table 1.5. Landform degree of activity.

Active or Quiescent	A
Inactive	I
Relict	R
Unknown	UN

IMP: If information on landform activity is unavailable (i.e., in the absence of geomorphological maps or field evidence), enter 'UN' or leave the field empty.

- New field: Description of geomorphic features

Insert a brief description of the landforms and the related processes that have shaped the landscape in the past and today.

- New field: Soil temperature regime

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Report the code of soil temperature regime according to table 1.6.

□

□

Annex 1 Explanation of soil temperature regimes (Adapted from USDA, 1999) (FAO, 2006)



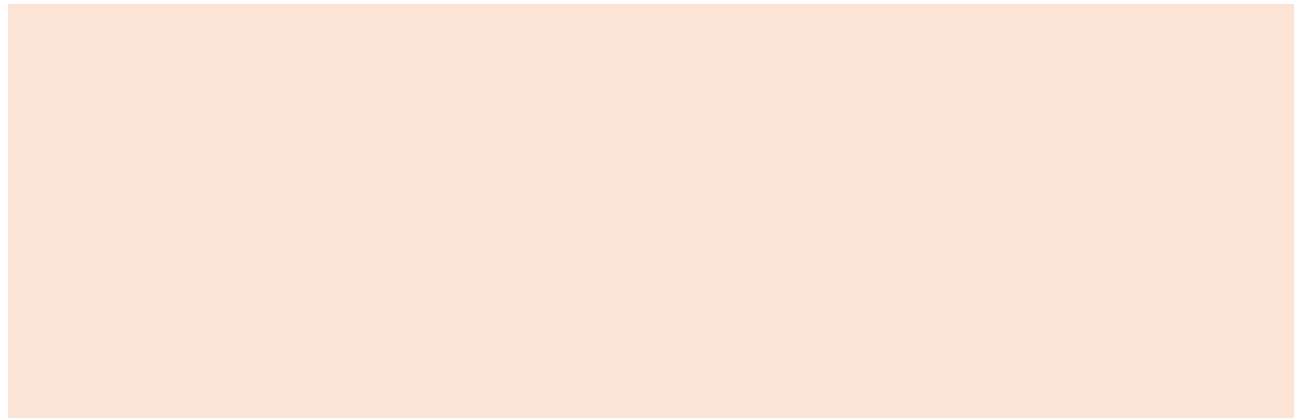


Table 1.6. Soil temperature regime codes, table 3 of guidelines for soil description (FAO, 2006)

Soil temperature regime			
PG	=	Pergelic	
CR	=	Cryic	
FR	=	Frigid	IF = Isofrigid
ME	=	Mesic	IM = Isomesic
TH	=	Thermic	IT = Isothermic
HT	=	Hyperthermic	IH = Isohyperthermic

- New field: Soil moisture regime

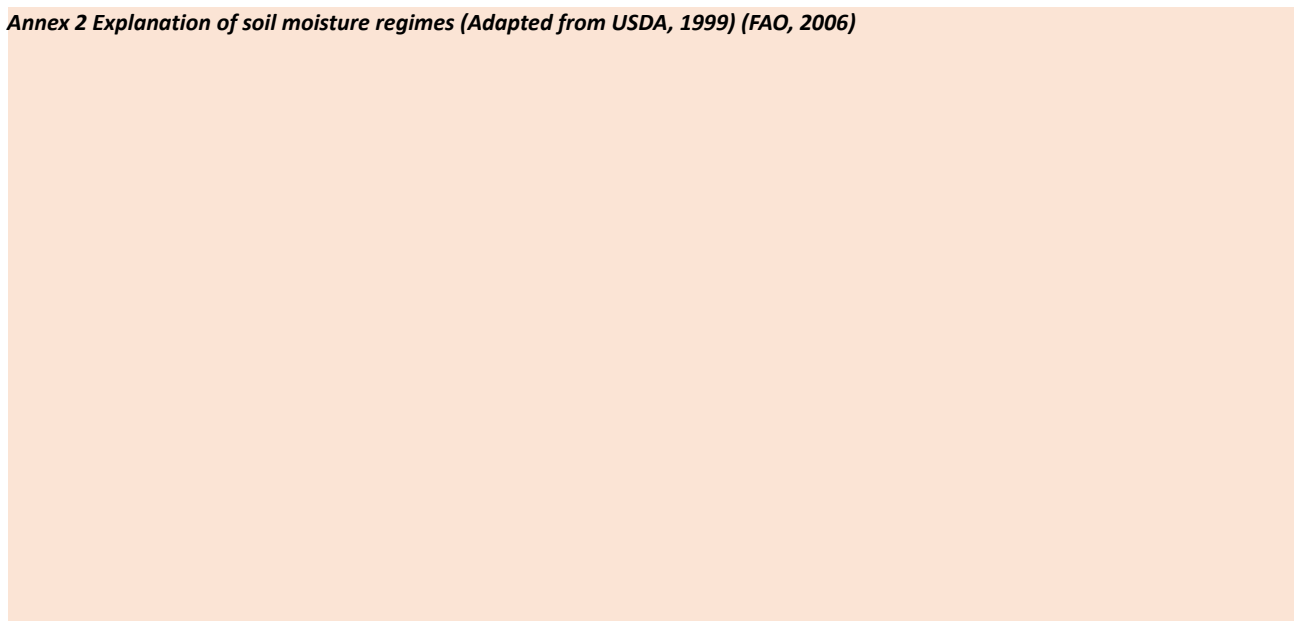
The field and its description are taken from the guidelines for soil description (FAO, 2006)

Report the code of soil temperature regime according to table 1.7.

□

□

Annex 2 Explanation of soil moisture regimes (Adapted from USDA, 1999) (FAO, 2006)





Land-use classification

A = Crop agriculture (cropping)		
AA	= Annual field cropping	
	AA1	= Shifting cultivation
	AA2	= Fallow system cultivation
	AA3	= Ley system cultivation
	AA4	= Rainfed arable cultivation
	AA5	= Wet rice cultivation
AP	AA6	= Irrigated cultivation
	= Perennial field cropping	
	AP1	= Non-Irrigated cultivation
AT	AP2	= Irrigated cultivation
	= Tree and shrub cropping	
	AT1	= Non-Irrigated tree crop cultivation
	AT2	= Irrigated tree crop cultivation
	AT3	= Non-Irrigated shrub crop cultivation
	AT4	= Irrigated shrub crop cultivation
M = Mixed farming		
MF	= Agroforestry	
MP	= Agropastoralism	
H = Animal husbandry		
HE	= Extensive grazing	
	HE1	= Nomadism
	HE2	= Semi-nomadism
	HE3	= Ranching
HI	= Intensive grazing	
	HI1	= Animal production
	HI2	= Dairying
F = Forestry		
FN	= Natural forest and woodland	
	FN1	= Selective felling
	FN2	= Clear felling
FP	= Plantation forestry	
P = Nature protection		
PN	= Nature and game preservation	
	PN1	= Reserves
	PN2	= Parks
	PN3	= Wildlife management
PD	= Degradation control	
	PD1	= Without interference
	PD2	= With interference
S = Settlement, industry		
SR	= Residential use	
SI	= Industrial use	
ST	= Transport	
SC	= Recreational use	
SX	= Excavations	
SD	= Disposal sites	
Y = Military area		
O = Other land uses		
LU = Not used and not managed		

- New field: Corine Land Cover Class

Report the “Corine Land Cover Class” (standard European land cover legend; last version available <https://land.copernicus.eu/content/corine-land-cover-nomenclature-guidelines/html/index.html>).

- New field: “Vegetation cover, by area of the ground cover [%]”, in Cultivated land

Report the vegetation cover. For the upper stratum and the mid-stratum, report the percentage (by area) of the crown cover. For the ground stratum, report the percentage (by area) of the ground cover.

Non-WRB Descriptors Bedrock

In this new sub-chapter the bedrock of the area should be described, reporting the lithology and the formation name, if available.

- New field: Geological Formation name

Report the name of the geological formation based on locally available maps or documents, if any.

- New field: Lithology

Report the lithology type according to the table 1.9.

If the type is unclear, or you do not find a match with the proposed ones, you can just use the general category (e.g. 'I' for Igneous Rock; etc.) or sub-category (e.g. 'IP' for Pyroclastic).

Table 1.9. Lithology type. Table modified from table 8.91 WRB 4TH EDITION (Annex 1).

Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	IF4
				Rhyolite	IF5
		Intermediate igneous	II	Andesite, trachyte, phonolite	II1
				Diorite-syenite	II2
		Mafic igneous	IM	Gabbro	IM1
				Basalt	IM2
				Dolerite	IM3
		Ultramafic igneous	IU	Peridotite	IU1
				Pyroxenite	IU2
				Serpentinite	IU3
		Pyroclastic	IP	Tuff, tuffite	IP1
				Volcanic scoria/breccia	IP2
				Volcanic ash	IP3
				Ignimbrite	IP4
Metamorphic rock	M	Felsic metamorphic	MF	Quartzite	MF1
				Gneiss, migmatite	MF2
				Slate, phyllite (pelitic rocks)	MF3
				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6
		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1
Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4

				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1
Sedimentary rock (unconsolidated)	U	Weathered residuum	UR	Bauxite, laterite	UR1
		Fluvial	UF	Sand and gravel	UF1
				Clay, silt and loam	UF2
		Lacustrine	UL	Sand	UL1
				Silt and clay, < 20% CaCO ₃ equivalent, little or no diatoms	UL2
				Silt and clay, < 20% CaCO ₃ equivalent, many diatoms	UL3
				Silt and clay, ≥ 20% CaCO ₃ equivalent (marl)	UL4
		Marine, estuarine	UM	Sand	UM1
				Clay and silt	UM2
		Colluvial	UC	Slope deposits	UC1
				Lahar	UC2
				Deposit of soil material	UC3
		Aeolian	UE	Loess	UE1
				Sand	UE2
		Glacial	UG	Moraine	UG1
				Glacio-fluvial sand	UG2
				Glacio-fluvial gravel	UG3
		Cryogenic	UK	Periglacial rock debris	UK1
				Periglacial solifluction layer	UK2
		Organic	UO	Rainwater-fed peat (bog)	UO1
				Groundwater-fed peat (fen)	UO2
				Lacustrine (organic limnic sediments)	UO3
		Anthropogenic/technogenic	UA	Redeposited natural material	UA1
				Industrial/artisanal deposits	UA2
		Unspecified deposits	UU	Clay	UU1
				Loam and silt	UU2
				Sand	UU3
				Gravelly sand	UU4
				Gravel, broken rock	UU5

- Modified field: “Size class” in Coarse surface fragments

The table 8.11 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting a new value of size class: “Gravel”. The code “Gravel” may be used if the data collected do not distinguish between fine, medium and coarse gravel.

Report the size of coarse fragments according to table 1.10.

Table 1.10. Size of coarse surface fragments, Modified version of Table 8.11 of the field guide WRB 4TH EDITION (Annex 1), FAO (2006), Table 15.

> 0.2 - 0.6	Fine gravel	F	Gravel	G
> 0.6 - 2	Medium gravel	M		
> 2 - 6	Coarse gravel	C		
> 6 - 20	Stones	S		
> 20 - 60	Boulders	B		
> 60	Large boulders	L		
No coarse surface fragments	N			

- New field: Drainage conditions

The field and its description are taken from the Soil Survey Manual (USDA, 2018).

Report the code of the soil drainage conditions class (Table 1.11).

Table 1.11. Drainage conditions class

excessively drained	ED
somewhat excessively drained	SED
well drained	WD
moderately well drained	MWD
Somewhat poorly drained	SPD
poorly drained	PD
very poorly drained	VPD
Subaqueous	SA

- New field: Depth of ground water (cm)

Report the depth of the ground water in cm.

- New field: Total area affected by erosion and deposition in Surface unevenness

The field and its description are taken from the guidelines for soil description (FAO, 2006).

Report the total area affected by erosion and deposition in %.

Non-WRB Descriptors: SOILS4MED Additional Descriptors

- New field: "Type of artefact" in Technogenic debris cover

Report the type of artefacts on the soil surface according to the table 1.12.

Table 1.12. Classification of kinds of artefacts, table 83 of guidelines for soil description (FAO, 2006).

Classification of kinds of artefacts

AN	Artesanal natural material
ID	Industrial dust
MM	Mixed material
OG	Organic garbage
PS	Pavements and paving stones
SL	Synthetic liquid
SS	Synthetic solid
WL	Waste liquid

- New field: "Surface cover (%)" in Technogenic debris cover

Report the overall artefact surface cover in %.

- New field: "Type" in Natural surface alterations

Report the type of natural surface alterations according to the table 1.13.

Table 1.13. Classification of type of natural surface alterations.

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disturbance by mammals	DM
mole's heaps	MH
self-mulching	SM

- New field: "Type" in Patterned compaction

Report the type of patterned compaction according to the table 1.14.

Table 1.14. Classification of type of patterned compaction.

Animal paths	AM
Grazing terracettes	GT
Technogenic paths	TP
Other	OTH

NB. Technogenic paths refer, for example, to traces left by mechanical vehicles, such as wheel ruts and track marks.

- New field: "Surface cover (%)" in Patterned compaction

Report the patterned compaction surface cover in %.

- New field: "Continuity" in Saline efflorescence on soil surface

Report the continuity of saline efflorescence on soil surface according to the table 1.15.

Leave the field blank if no efflorescence is observed.

Table 1.15. Classification of continuity of saline efflorescence on soil surface.

Continuos	CN
Discontinuos	DS

- New field: "Abundance" in Worm cast

Count the average number of worm cast per m². Write 0 if you don't see any.

Non-WRB Descriptors: SOILS4AFRICA Additional Descriptors

These new fields are taken from the SOILS4AFRICA Protocols of field survey version 3.0.

For more details see SOILS4AFRICA manual chapter 8.3.1 for Land preparation, chapter 8.3.2 for Use of inputs, chapter 8.4 for Irrigation type and chapter 8.5 for Conservation measure type.

- New field: "Sign of ploughing" in Land preparation

Reports the sign of ploughing according to table 1.16

Table 1.16. Signs of ploughing/tillage. Modified version of Table in chapter 8.3.1 Land preparation of the SOILS4AFRICA Protocols of field survey version 3.0.

No visible signs of tillage	0
No-till	1
Signs of tillage in the past: ridges visible forming a pattern, but ridges less pronounced/flattened	2
Signs of field recently tilled: patterns of ridges clearly visible that form a pattern	3

- New field: "Direction of ploughing" in Land preparation

Reports the direction of ploughing according to table 1.17.

Table 1.17. Direction of ploughing. Modified version of Table in chapter 8.3.1 Land preparation of the SOILS4AFRICA Protocols of field survey version 3.0.

Not applicable	0
Along contours	1
In slope direction, across contour	2

- New field: "Tillage mode" in Land preparation

Reports the tillage mode according to table 1.18.

Table 1.18. Tillage mode. Modified version of Table in chapter 8.3.1 Land preparation of the SOILS4AFRICA Protocols of field survey version 3.0.

Not applicable	0
Manual (hoe)	1
Animal traction (buffalo, bullock, cow, horse, mule/donkey)	2
Mechanical (tractor)	3

- New field: "Use of inputs"

Reports the use of inputs according to table 1.19.

Table 1.19. Use of inputs. Modified version of Table in chapter 8.3.2 Use of inputs of the SOILS4AFRICA Protocols of field survey version 3.0.

No visible signs	0
Crop residues	1
Green manures and/or compost	2
Manure (FYM, cattle manure, chicken manure, etc.)	3
Inorganic fertilizer	4

Enter a maximum of four types of signs of use of input per cell separated by commas (e.g. 1,2,3,4)

- New field: "Irrigation type"

Reports the irrigation type according to table 1.20.

Table 1.20. Irrigation type. Modified version of Table in chapter 8.4 Water management/ irrigation of the SOILS4AFRICA Protocols of field survey version 3.0.

Surface / gravity (can be by use borders, basins, furrow, corrugation (i.e., parallel ridges and grooves, wild flooding, etc.)	1
Sprinkler (different types of sprinklers: central pivot, fixed set of sprinklers, travelling gun irrigation system, etc.)	2
Drip irrigation (trickle, dribble, or localized irrigation in which the water trickles onto or into the soil near the plant)	3
Not applicable	4

- New field: "Conservation measure type"

Reports the conservation measure type according to table 1.21.

Table 1.21. Conservation measure type. Modified version of Table in chapter 8.5 Soil and water conservation of the SOILS4AFRICA Protocols of field survey version 3.0.

Stone line	1

Bench terrace	2
Graded terrace	3
Contour bunds	4
Graded bunch	5
Drains, ditches, and furrows (for retention of water and/or soil)	6
Grass strips (vegetated strips)	7
Not applicable	8

☐ New field: "Notes"

Report in this field the important information that cannot be inserted in any previous field but could be useful.

Non-WRB Descriptors Soil classification

This chapter has been inserted to collect the soil classification info of the profiles in the database.

- "Old classification"

Report the original classification of the soil profile.

- Classification system

Report the original classification system used to classify the soil according to Table 1.22.

Table 1.22. Classification system.

USDA	US	1960	US60
		1975	US75
		1990	US90
		1991	US91
		1994	US94
		1998	US98
		1999	US99
		2006	US06
		2010	US10
		2014	US14
		2022	US22
FAO	FA	1967	FA67
		1974	FA74
		1985	FA85
		1988	FA88
		1994	FA94
WRB	WR	1998	WR98
		2006	WR06
		2015	WR15
		2022	WR22
CPCS	CP		
Referentiel Pedologique	RP	1995	RP95
		2008	RP08
KUBIENA	KU		
OTHER	OTH		
UNKNOWN	UN		

- New classification (WRB 4TH EDITION)

Report the new soil name based on the WRB 4TH EDITION.

The translation from the different reference systems used to WRB 4TH EDITION will be done on the basis of conversion tables, which will be provided later.

- New field: “Point ID”

Coincides with the Point ID defined in Chapter 8.2.1

☑ Use the same code inserted in the field “Point ID” of sheet 1.

- Moved field: Horizon designation

For convenience the field is now at the beginning of the table, while it was at the end in the official version.

When translating horizon names from other classification systems, particular attention shall be paid to the “Oi” horizon. It should be assessed whether to retain the same designation or to report the information in the litter field of the “General and surface” sheet.

- Field: Lower boundary

Enter only the numeric value and no symbol (e.g. '+').

If the lower limit has not been reached, leave the field blank and enter 'U = unknown ' in the column “Distinctness of the layer's lower boundary”.

- New Field: “lower boundary range”

As explained in FAO (2006) “Most horizons do not have a constant depth. The variation or irregularity of the surface of the boundary is described by the topography in terms of smooth, wavy, irregular and broken. If required, ranges in depth should be given in addition to the average depth, for example 28 (25 to 31) cm or 45 (39 to 51) cm. In this case insert in this field the range of depth (e.g., besides 28, input “25-31”).

- New field: “Layer thickness”

Report the thickness value of the different layers.

If not filled in, the thickness value will be inserted automatically based on the lower boundary data.

- Modified field: “Size and shape” in Coarse fragments

The table 8.33 in the WRB 4TH EDITION field guide (Annex 1) has been modified inserting two new values of shape for each size class: “Subrounded” and “Flat” and inserting a code only for size classes according to the Table 27 and 28 of FAO (2006).

The general category “Gravel” is also inserted. Use this category if it is not possible to divide the gravel into size sub-groups.

Report the size and shape of coarse material according to table 2.1.

Table 2.1. Size and shape classes of coarse fragments Modified version of Table 8.33 of the field guide WRB 4TH EDITION (Annex 1), Tables 27 and 28, FAO (2006),

> 0.2 - 6	Gravel	G	Rounded	GR
			Angular	GA

			Rounded and angular	GB
			Subrounded	GS
			Flat	GF
> 0.2 - 0.6	Fine gravel	F	Rounded	FR
			Angular	FA
			Rounded and angular	FB
			Subrounded	FS
			Flat	FF
> 0.6 - 2	Medium gravel	M	Rounded	MR
			Angular	MA
			Rounded and angular	MB
			Subrounded	MS
			Flat	MF
> 2 - 6	Coarse gravel	C	Rounded	CR
			Angular	CA
			Rounded and angular	CB
			Subrounded	CS
			Flat	CF
> 6 - 20	Stones	S	Rounded	SR
			Angular	SA
			Rounded and angular	SB
			Subrounded	SS
			Flat	SF
> 20 - 60	Boulders	B	Rounded	BR
			Angular	BA
			Rounded and angular	BB
			Subrounded	BS
			Flat	BF
> 60	Large boulders	L	Rounded	LR
			Angular	LA
			Rounded and angular	LB
			Subrounded	LS
			Flat	LF
None		NO		

- New fields: “Lithology of size and shape class 1”, “Lithology of size and shape class 2”, “Lithology of size and shape class 3”, “Lithology of size and shape class 4” in Coarse fragments

Report the lithology of the coarse fragments according to the table 2.2.

Table 2.2. Lithology of coarse fragments, modified version of table 8.91 of the field guide WRB 4TH EDITION (Annex 1) (Modified FAO, 2006). Enter a maximum of three lithology types per cell separated by commas (e.g. IM,SE,MF)

Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	IF4
				Rhyolite	IF5
		Intermediate igneous	II	Andesite, trachyte, phonolite	II1
				Diorite-syenite	II2
		Mafic igneous	IM	Gabbro	IM1
				Basalt	IM2
				Dolerite	IM3
		Ultramafic igneous	IU	Peridotite	IU1
				Pyroxenite	IU2
				Serpentinite	IU3
		Pyroclastic	IP	Tuff, tuffite	IP1
				Volcanic scoria/breccia	IP2
				Volcanic ash	IP3
				Ignimbrite	IP4
Metamorphic rock	M	Felsic metamorphic	MF	Quartzite	MF1
				Gneiss, migmatite	MF2
				Slate, phyllite (pelitic rocks)	MF3
				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6
		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1
Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4
				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1

- Modified field: “Cementing agent” in Remnants of broken-up cemented layers

The table 8.35 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting a general code “F” for Fe oxides cementing agent. Report the cementation agent according to table 2.3.

Table 2.3. Remnants of broken-up cemented layers: cementing agent, Modified version of Table 8.35 of the field guide WRB 4TH EDITION (Annex 1).

Secondary carbonates		CA
Secondary gypsum		GY
Secondary silica		SI
Fe oxides, predominantly inside (former) soil aggregates, no significant concentration of organic matter	F	FI
Fe oxides, predominantly on the surfaces of (former) soil aggregates, no significant concentration of organic matter		FO
Fe oxides, no relationship to (former) soil aggregates, no significant concentration of organic matter		FN
Fe oxides in the presence of a significant concentration of organic matter		FH

- Modified field: “Type” in Structure

The table 8.42 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting two new code of structure type: “Single grain” and “Massive”. The code “Single grain” or “Massive” may be used if the layer does not show pedogenic structure. Report the structure according to table 2.4.

Table 2.4. Types of structure, formation and codes, Modified version of Table 8.42 of the field guide WRB 4TH EDITION (Annex 1).

Granular	Soil aggregate structure, natural	GR		
Subangular blocky	Soil aggregate structure, natural	BS		
Angular blocky	Soil aggregate structure, natural	BA		
Lenticular	Soil aggregate structure, natural	LC		
Wedge-shaped	Soil aggregate structure, natural	WE		
Prismatic	Soil aggregate structure, natural	PR		
Columnar	Soil aggregate structure, natural	CO		
Polyhedral	Soil aggregate structure, natural	PH		
Flat-edged	Soil aggregate structure, natural	FE		
Pseudosand/Pseudosilt	Soil aggregate structure, natural	PS		
Platy	Soil aggregate structure, natural or resulting from artificial pressure	PL		
Single grain	No structural units	S	No structural units, rock structure, inherited from the parent material	SR
			No structural units, soil structure, resulting from	SS

			soil-forming processes, like loss of organic matter and/or oxides and/or clay minerals or loss of stratification	
Massive	No structural units	M	No structural units, rock structure, inherited from the parent material, structure not changing with soil moisture, not or only slightly chemically weathered	MR
			No structural units, rock structure, inherited from the parent material, structure not changing with soil moisture, strongly chemically weathered (e.g. saprolite)	MW
			No structural units, soil structure, present when moist and changing into soil aggregate structure when dry	MS
Stratified	No structural units, rock structure, visible stratification from sedimentation	ST		
Cloddy	Artificial structural elements	CL		

- Modified field: “Type” in Non matrix pores

A new Type “Other” and related criterion have been inserted in the table, in order to allow the description of those pores not included in the types listed above.

Table 2.5. Types of non-matrix pores, modified version of Table 8.46 of the field guide WRB 4TH EDITION (Annex 1), Schoeneberger et al. (2012), 2-73, modified

Cylindrical and elongated voids; e.g., worm tunnels	Tubular	TU
Cylindrical, elongated, branching voids; e.g., empty root channels	Dendritic Tubular	DT
Ovoid to spherical voids; e.g., solidified pseudomorphs of entrapped gas bubbles concentrated below a crust; most common in arid and semiarid environments and in permafrost soils	Vesicular	VE
Non-connected cavities, chambers; e.g., vughs; various shapes	Irregular	IG
Pores not attributable to any of the previous classes	Other	OT
No non-matrix pores		NO

- Modified field: “Location” in Redoximorphic features

A new Location “Mottles” has been inserted in Table 8.55, Location to select in case of presence of mottles in the layer.

Table 2.6. Location of oximorphic features. Modified version of Table 8.55 of the field guide WRB 4TH EDITION (Annex 1).

Inner parts	OI	Inside soil aggregates: masses	OIM
		Inside soil aggregates: concretions	OIC
		Inside soil aggregates: nodules	OIN
		Inside soil aggregates: both concretions and/or nodules (not possible to distinguish)	OIB
Outer parts	OO	On surfaces of soil aggregates	OOA
		Adjacent to surfaces of soil aggregates, infused into the matrix (hypocoats)	OOH
		On biopore walls, lining the entire wall surface	OOE
		On biopore walls, not lining the entire wall surface	OON
		Adjacent to biopores, infused into the matrix (hypocoats)	OOI
Random (not associated with aggregate surfaces or pores)	OR	Distributed over the layer, no order visible	ORN
		Distributed over the layer, surrounding areas with reductimorphic features	ORS
		Throughout	ORT
Mottles	OM		

- New field: “Contrast of mottles” in Redoximorphic features

Report the contrast of mottles if mottles have been selected in the location field according to the table 2.7. The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.7. Classification of the contrast of mottles, table 34 of guidelines for soil description (FAO, 2006)

Classification of the contrast of mottles	
F Faint	The mottles are evident only on close examination. Soil colours in both the matrix and mottles have closely related hues, chromas and values.
D Distinct	Although not striking, the mottles are readily seen. The hue, chroma and value of the matrix are easily distinguished from those of the mottles. They may vary by as much as 2.5 units of hue or several units in chroma or value.
P Prominent	The mottles are conspicuous and mottling is one of the outstanding features of the horizon. Hue, chroma and value alone or in combination are at least several units apart.

- New field: “Boundary between mottles and matrix” in Redoximorphic features

Report the characteristics of boundary between mottles and matrix if mottles have been selected in the location field, according to the table 2.8.

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.8. Classification of boundary between mottle and matrix, table 35 of guidelines for soil description (FAO, 2006)

Classification of boundary between mottle and matrix		
		mm
S	Sharp	< 0.5
C	Clear	0.5–2
D	Diffuse	> 2

- New field: “Form of coatings” in Coatings and bridges

Report the form for each category of coatings according to the table 2.9.

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.9. Classification of the form of coatings, table 67 of guidelines for soil description (FAO, 2006)

Classification of the form of coatings	
C	Continuous
CI	Continuous Irregular (non-uniform, heterogeneous)
DI	Discontinuous Irregular
DE	Dendroidal
DC	Discontinuous circular
O	Other

- New fields: “Size” and “Shape” in Carbonates

Report the size and shape for each type of secondary carbonates (i.e., “Size of Type 1”, “Shape of Type 1”, “Size of Type 2”, “Shape of Type 2”, “Size of Type 3”, “Shape of Type 3”, “Size of Type 4”, “Shape of Type 4”) according to the table 2.10.

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.10. Classification of the size and shape of mineral concentrations, table 75 of guidelines for soil description (FAO, 2006)

Classification of the size and shape of mineral concentrations

	Size	(mm)	Shape
V	Very fine	< 2	R Rounded (spherical)
F	Fine	2–6	E Elongated
M	Medium	6–20	F Flat
C	Coarse	> 20	I Irregular
			A Angular

- New field: “Type” (Type 1 and Type 2) in Gypsum

Report the type of secondary gypsum according to the table 2.11.

Table 2.11. Type of form of secondary gypsum.

Gypsum crystal intergrowths or nodules	CR
Soft concretions	SC
Disperse powdery gypsum	DP
Filaments (vermiform gypsum, pseudomycelia)	FL
Other	OT

- New field: “Size of Type 1”, “Shape of Type 1”, “Size of Type 2” and “Shape of Type 2” in Gypsum

Report the size and shape of each type according to the table 2.12.

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.12. Classification of the size and shape of mineral concentrations, table 75 of guidelines for soil description (FAO, 2006)

Classification of the size and shape of mineral concentrations

	Size	(mm)	Shape
V	Very fine	< 2	R Rounded (spherical)
F	Fine	2–6	E Elongated
M	Medium	6–20	F Flat
C	Coarse	> 20	I Irregular
			A Angular

- New field: “Continuity” in Saline efflorescence

Report the continuity of saline efflorescence according to the table 2.13.

Leave the field blank if no efflorescence is observed.

Table 2.13. Classification of continuity of saline efflorescence.

Continuos	CN
Discontinuos	DS

- New field: “Stickiness” in Consistence

Report the stickiness of each horizon, according to the table 2.14.

The field and its description are taken from the guidelines for soil description (FAO, 2006)

Table 2.14. Classification of soil stickiness, table 55 of guidelines for soil description (FAO, 2006)

Classification of soil stickiness		
NST	Non-sticky	After release of pressure, practically no soil material adheres to thumb and finger.
SST	Slightly sticky	After pressure, soil material adheres to both thumb and finger but comes off one or the other rather cleanly. It is not appreciably stretched when the digits are separated.
ST	Sticky	After pressure, soil material adheres to both thumb and finger and tends to stretch somewhat and pull apart rather than pulling free from either digit.
VST	Very sticky	After pressure, soil material adheres strongly to both thumb and finger and is decidedly stretched when they are separated.

- New field: “Average thickness (cm)” in Surface crusts

Report the average thickness of the surface crusts in cm.

- New fields: “Abundance of very fine roots (<0.5 mm)” and “Abundance of fine roots (0.5-2 mm)”;
“Abundance of medium roots (2-5 mm)” and “Abundance of coarse roots (>5 mm)”, in Roots.

The diameter roots subdivision is made according to the table 18. 79 (FAO, 2006).

Table 2.15. Classification of the diameter of roots, table 78 of guidelines for soil description (FAO, 2006)

Classification of the diameter of roots		
		mm
VF	Very fine	< 0.5
F	Fine	0.5–2
M	Medium	2–5
C	Coarse	> 5

Note: Additional codes are: FF, very fine and fine; FM, fine and medium; and MC, medium and coarse.

Report the abundance for each groups of roots according to Table 8.86: Abundance of roots, WRB 4TH EDITION (Annex 1)

- Modified field: Parent Material

Report the parent material according table 2.16.

Table 2.16. Lithology type. Table modified from table 8.91 WRB 4TH EDITION (Annex 1).

Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	IF4
				Rhyolite	IF5
		Intermediate igneous	II	Andesite, trachyte, phonolite	II1
				Diorite-syenite	II2
		Mafic igneous	IM	Gabbro	IM1
				Basalt	IM2
				Dolerite	IM3
		Ultramafic igneous	IU	Peridotite	IU1
				Pyroxenite	IU2
				Serpentinite	IU3
		Pyroclastic	IP	Tuff, tuffite	IP1
				Volcanic scoria/breccia	IP2
				Volcanic ash	IP3
				Ignimbrite	IP4
Metamorphic rock	M	Felsic metamorphic	MF	Quartzite	MF1
				Gneiss, migmatite	MF2
				Slate, phyllite (pelitic rocks)	MF3
				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6

		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1
Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4
				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1
Sedimentary rock (unconsolidated)	U	Weathered residuum	UR	Bauxite, laterite	UR1
		Fluvial	UF	Sand and gravel	UF1
				Clay, silt and loam	UF2
		Lacustrine	UL	Sand	UL1
				Silt and clay, < 20% CaCO ₃ equivalent, little or no diatoms	UL2
				Silt and clay, < 20% CaCO ₃ equivalent, many diatoms	UL3
				Silt and clay, ≥ 20% CaCO ₃ equivalent (marl)	UL4
		Marine, estuarine	UM	Sand	UM1
				Clay and silt	UM2
		Colluvial	UC	Slope deposits	UC1
				Lahar	UC2
				Deposit of soil material	UC3
		Aeolian	UE	Loess	UE1
				Sand	UE2
		Glacial	UG	Moraine	UG1
				Glacio-fluvial sand	UG2
				Glacio-fluvial gravel	UG3
		Cryogenic	UK	Periglacial rock debris	UK1
				Periglacial solifluction layer	UK2
		Organic	UO	Rainwater-fed peat (bog)	UO1
				Groundwater-fed peat (fen)	UO2
				Lacustrine (organic limnic sediments)	UO3

		Anthropogenic/ technogenic	UA	Redeposited natural material	UA1
				Industrial/artisanal deposits	UA2
		Unspecified deposits	UU	Clay	UU1
				Loam and silt	UU2
				Sand	UU3
				Gravelly sand	UU4
				Gravel, broken rock	UU5

New field: "Notes"

Report in this field the important information that cannot be inserted in any previous field but could be useful.

The "Lab data" and "Lab data by sampling depth" sheets constitute the data model for laboratory-based determinations. They are identical, with the exception of the first three columns.

Use the sheet to enter laboratory data related to soil samples collected from, and corresponding to, the genetic horizons described in Sheet 2 (thus perfectly associated with them). Of course, there may be horizons for which no sample was collected.

In this case, each sample is identified by the Point ID, the Horizon designation, and the Number of the layer.

Use the sheet to enter laboratory data related to soil samples collected according to depth layers. These samples correspond to the layers described in Sheet 2 only if also those layers, unlike genetic horizons, have been distinguished and described according to the same depth layers (in this case samples are perfectly associated to layers). On the contrary, if genetic horizons are described in Sheet 2 but samples were collected according to depth, the sample data of Sheet 3 will not be perfectly associated with them. Of course, in this case one may have multiple thin horizons described in Sheet 2, and only one or two depth layers sampled in Sheet 3.

In this case, each sample is identified by the Point ID and by the upper lower boundaries that indicate the sampling depth.

The data fields are divided into two types:

- Fields for the of the parameter
- Fields for the used to obtain the data

The method used must be selected from those proposed in the table associated with the field.

If the method used is unknown, enter "UN", i.e. unknown, and if the method used is not one of the options in the table, enter "OTH", i.e. other, and specify the method used in the notes field.

When entering data, pay attention to the used for each field.

In the case of data expressed in different units, a conversion must be carried out before inserting the data.

- Point ID

Coincides with the Point ID defined in Chapter 8.2.1

☐ Use the same code inserted in the field “Point ID” of sheet 1.

- Horizon designation

☐ Use the same designation inserted in the field “Horizon designation” of Sheet 2 - “Layer description”.

- Number of the layer

☐ Use the same number inserted in the field “Number of the layer” of Sheet 2 - “Layer description”.

- Sampling upper boundary and Sampling lower boundary

There are two possible cases:

1) If the depth layer used for sampling corresponds to the layer boundaries indicated in Sheet 2, then:

☐ Insert the same value in the fields “Upper boundary” and “Lower boundary” of “Sheet 2 - “Layer description”.

2) If, like in the case of the soil health monitoring data generated by SOILS4MED project, the sampling depth layer does not correspond to the layer boundaries indicated in Sheet 2, then:

☐ Insert the actual upper and lower value of the sampling depth layer

Note for SOILS4MED partners.

We may have three operation cases during the soil health monitoring work.

Case 1, standard: Indicate 0-20 and 20-50 depth layers

oring

WRB	W
OTHER	OT
UNKNOWN	UN

- Sand

Enter the content of sand in % on the fine earth.

If the data you have does not match any of the sand classes listed in the fields below, simply enter the total sand value and specify the subclass types in the 'Notes' field.

- Very coarse sand

Enter the content of very coarse sand in % on the fine earth.

- Coarse sand

Enter the content of coarse sand in % on the fine earth.

- Medium sand

Enter the content of medium sand in % on the fine earth.

- Fine sand

Enter the content of fine sand in % on the fine earth.

- Very fine sand

Enter the content of very fine sand in % on the fine earth.

- Method used for sand content

Report the methodology used according to table 3.2.

Table 3.2.

Sieving	SV
hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Silt

Enter the content of silt in % on the fine earth

If the data you have does not match any of the silt classes listed in the fields below, simply enter the total sand value and specify the subclass types in the 'Notes' field.

- Coarse silt

Enter the content of coarse silt in % on the fine earth.

- Fine silt

Enter the content of fine silt in % on the fine earth.

- Method used for silt content

Report the type of methodology used according to table 3.3.

Table 3.3.

hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Clay

Enter the content of clay in % on the fine earth.

- Method used for clay content

Report the type of methodology used according to table 3.4.

Table 3.4.

hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Texture class

Report the type of texture class type according to table 3.5.

Table 3.5. Table 8.39 of the field guide WRB 4TH EDITION (Annex 1), Soil Science Division Staff (2017)

Sand (S)	S
Loamy sand	LS
Silt	Si
Silt loam	SiL
Sandy loam	SL
Loam	L
Sandy clay loam	SCL
Silty clay loam	SiCL
Clay loam	CL

Sandy clay	SC
Silty clay	SiC
Clay	C

- Bulk density

Enter the bulk density value in g/cm³.

- Method used for bulk density

Report the type of methodology used according to table 3.6.

Table 3.6.

Core method	CM
Excavation method	EM
Clod method	LM
Other	OTH
Unknown	UN

- Slake test

Enter the slake test value about aggregate stability.

- Method used for slake test

Report the type of methodology used according to table 3.7.

Table 3.7.

Smartphone “Slakes” application form the Soil Health Institute	APS
Other	OTH
Unknown	UN

PS. The smartphone “Slakes” application form the Soil Health Institute is available at <https://soilhealthinstitute.org/our-work/initiatives/slakes/>

- Electrical conductivity

Enter the electrical conductivity value in dS/m.

- Method used for electrical conductivity

Report the type of methodology used according to table 3.8.

Table 3.8.

Saturated paste	SP
Soil to water 1:2.5	OP

Soil to water 1:5	OF
Soil to water 1:1	OO
Soil to water 1:2	OT
Other	OTH
Unknown	UN

- Hydraulic conductivity at saturation (Ksat)

Enter the value of hydraulic conductivity at saturation in mm/h.

- Method used for Hydraulic conductivity at saturation (Ksat)

Report the type of methodology used according to table 3.9.

Table 3.9.

Infiltrometer	IN
Constant pressure	CP
Pressure drop	PD
Hole - auger	HA
Double tube	DT
Other	OTH
Unknown	UN

- Water Saturation

Enter the water saturation value in %.

- Field capacity

Enter the field capacity value in %.

- Wilting point

Enter the wilting point value in %.

- AWC

Enter the Available Water Capacity value in %.

- Method used for water saturation, field capacity, wilting point

Report the type of methodology used according to table 3.10.

Table 3.10.

Pressure Plate Method	PP

Other	OTH
Unknown	UN

- Exchangeable acidity

Enter the exchangeable acidity value in cmol/Kg.

- Method used for Exchangeable Acidity

Report the type of methodology used.

- pH (H₂O)

Enter the pH (H₂O) value.

- Method used for pH (H₂O)

Report the type of methodology used according to table 3.11.

Table 3.11.

Soil water or salt solution ratio 1:1	OO
Soil water or salt solution ratio 1:2.5	OP
Soil water or salt solution ratio 1:5	OF
Other	OTH
Unknown	UN

- pH (KCl)

Enter the pH (KCl) value.

- Method used for pH (KCl)

Report the type of methodology used according to table 3.11.

- pH (CaCl₂)

Enter the pH (CaCl₂) value.

- Method used for pH (CaCl₂)

Report the type of methodology used according to table 3.11.

- Organic Carbon content

Enter the organic carbon content in g/Kg.

- Method used for Organic Carbon content

Report the type of methodology used according to table 3.12.

Table 3.12.

Walkley –Black	WB

Elemental analyzer	EA
Springer & Klee	SK
Other	OTH
Unknown	UN

- Organic matter content

Enter the organic matter content in %.

- Method used for organic matter content

Report the type of methodology used according to table 3.13.

Table 3.13.

Muffle	MF
Other	OTH
Unknown	UN

- CaCO₃ content

Enter the calcium carbonate content in %.

- Method used for CaCO₃ content

Report the type of methodology used according to table 3.14.

Table 3.14.

Calcimeter - Gas volumetric method	GV
Carbonate equivalent, Gravimetric method	GM
Titration method	TM
Other	OTH
Unknown	UN

- Active CaCO₃

Enter the active calcium carbonate content in %.

- Method used for Active CaCO₃

Report the type of methodology used according to table 3.15.

Table 3.15.

Droineau method	DM
Other	OTH

Unknown	UN
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- Gypsum content

Enter the gypsum content in %.

- Method used for Gypsum content

Report the type of methodology used according to table 3.16.

Table 3.16.

Acetone, EDTA titration (George Holmgren)	AT
Soluble Calcium, FAAS	FA
Other	OTH
Unknown	UN

- CEC

Enter the value of cation exchange capacity in cmol/Kg.

- Method used for CEC

Report the type of methodology used according to table 3.17.

Table 3.17.

BaCl ₂ /MgSO ₄ method	BM
Ammonium Acetate (CH ₃ COONH ₄)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- Exch. Ca⁺⁺

Enter the value of exchangeable Ca⁺⁺ in cmol/Kg.

- Exch. Mg⁺⁺

Enter the value of exchangeable Mg⁺⁺ in cmol/Kg.

- Exch. Na⁺

Enter the value of exchangeable Na⁺ in cmol/Kg.

- Exch. K⁺

Enter the value of exchangeable K⁺ in cmol/Kg.

- Method used for exchangeable cations

Report the type of methodology used according to table 3.18.

Table 3.18.

Ammonium Acetate (CH ₃ COONH ₄)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- Base saturation

Enter the value of base saturation in %.

- ESP

Enter the value of exchangeable sodium percentage.

- Sol. Ca⁺⁺

Enter the value of soluble Ca⁺⁺ in cmol/L.

- Sol. Mg⁺⁺

Enter the value of soluble Mg⁺⁺ in cmol/L.

- Sol. Na⁺

Enter the value of soluble Na⁺ in cmol/L.

- Method used for soluble cations

Report the type of methodology used according to table 3.19.

Table 3.19.

Saturated soil extract	SSE
Other	OTH
Unknown	UN

- SAR

Enter the sodium adsorption ratio.

- N tot content

Enter the total nitrogen content in g/Kg.

- Method used for N tot content

Report the type of methodology used according to table 3.20.

Table 3.20.

Kjeldahl	KJ
Elemental analyzer	EA

Coleman nitrogen analyzer	CA
Microwave	MW
Other	OTH
Unknown	UN

- Available P content

Enter the available phosphorus content in mg/kg.

- Method used for available P content

Report the type of methodology used according to table 3.21.

Table 3.21.

Olsen	OL
Bray	BR
Mehlich No. 3	MN
Other	OTH
Unknown	UN

- NH₄⁺

Enter the ammonium content in mg/Kg.

- Method used for NH₄⁺

Report the type of methodology used according to table 3.22.

Table 3.22.

KCl 1M extract	K1E
KCl 2M extract	K2E
Other	OTH
Unknown	UN

- NO₃⁻

Enter the nitrate content in mg/Kg.

- Method used for NO₃⁻

Report the type of methodology used according to table 3.23.

Table 3.23.

KCl 1M extract	K1E

KCl 2M extract	K2E
Water extract	WE
Other	OTH
Unknown	UN

- ROC

Enter the residual oxidizable carbon content in mg/Kg.

- TOC400

Enter the thermally labile organic carbon content in g/Kg.

- Method used for ROC and TOC400

Report the type of methodology used according to table 3.24.
Table 3.24.

LECO RC612 analyser	LEC
Other	OTH
Unknown	UN

- Fe_{ox}

Enter the value of oxalate-extractable iron in g/kg.

- Fe_d

Enter the value of dithionite-extractable iron in g/kg.

- Fe_p

Enter the value of pyrophosphate-extractable iron in g/kg.

- Fe_{tot}

Enter the value of total iron in g/kg.

- Method used for Fe_{tot}

Report the type of methodology used according to table 3.25.
Table 3.25.

Acqua Regia	AR
Fluoridric Acid	FA
XRF	XRF
Other	OTH
Unknown	UN

- Mn

Enter the manganese content in mg/kg.

- Method used for Mn

Report the type of methodology used according to table 3.26.

Table 3.26.

Nitric acid, FAAS	NAF
EDTA, FAAS	EF
DTPA, FAAS	DF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
EDTA, ICP-MS	EIM
DTPA, ICP-MS	DIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
EDTA, ICP-AES	EIA
DTPA, ICP-AES	DIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Zn

Enter the zinc content in mg/kg.

- Method used for Zn

Report the type of methodology used according to table 3.27.

Table 3.27.

Nitric acid, FAAS	NAF
EDTA, FAAS	EF
DTPA, FAAS	DF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM

EDTA, ICP-MS	EIM
DTPA, ICP-MS	DIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
EDTA, ICP-AES	EIA
DTPA, ICP-AES	DIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Cu

Enter the copper content in mg/kg.

- Method used for Cu

Report the type of methodology used according to table 3.28.

Table 3.28.

Nitric acid, FAAS	NAF
EDTA, FAAS	EF
DTPA, FAAS	DF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
EDTA, ICP-MS	EIM
DTPA, ICP-MS	DIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
EDTA, ICP-AES	EIA
DTPA, ICP-AES	DIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Pb

Enter the lead content in mg/kg.

- Method used for Pb

Report the type of methodology used according to table 3.29.

Table 3.29.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Hg

Enter the mercury content in mg/kg.

- Method used for Hg

Report the type of methodology used according to table 3.30.

Table 3.30.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Cd

Enter the cadmium content in mg/kg.

- Method used for Cd

Report the type of methodology used according to table 3.31.

Table 3.31.

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Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid,	

- Cr

Enter the chromium content in mg/kg.

- Method used for Cr

Report the type of methodology used according to table 3.34.

Table 3.34.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- As

Enter the arsenic content in mg/kg.

- Method used for As

Report the type of methodology used according to table 3.35.

Table 3.35.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- Co

Enter the cobalt content in mg/kg.

- Method used for Co

Report the type of methodology used according to table 3.36.

Table 3.36.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

- V

Enter the vanadium content in mg/kg.

- Method used for V

Report the type of methodology used according to table 3.37.

Table 3.37.

Nitric acid, FAAS	NAF
DTPA ammonium bicarbonate, FAAS	DBF
Nitric acid, ICP-MS	NAIM
DTPA ammonium bicarbonate, ICP-MS	DBIM
Nitric acid, ICP-AES	NAIA
DTPA ammonium bicarbonate, ICP-AES	DBIA
Other	OTH
Unknown	UN

Notes

Report in this field the important information that cannot be inserted in any previous field but could be useful.

For each dataset, the Excel file relating to the genealogy of the data must also be complete (S4M_data_genealogy_template_v2).

This “Data genealogy” spreadsheet consists of a single sheet with the following fields: “Project ID”, “Project title”, “Project description”, “References”, “Year of publication”, “Average accuracy of point location coordinates”, “Web link”, “Data owners” and “Data access restrictions”.

A brief description of each field can be found below.

- Project ID

☐ Use the same code inserted in the field “Project ID” of Sheet 1 of the point soil data .xlsx template.

- Project title and original ID

Enter the Acronym and full Title of the Project (in case of structured projects), or the name of the Survey activity/campaign.

- Project description

Report a brief description of the project or Survey activity/campaign, especially summarizing the main features of the activities/surveys that generated the soil point data.

- References

Enter the references, in the form of bibliographic reference, indicating the dataset authors, date, and name. Also add references of data papers, or documents illustrating the soil datasets, if any.

- Year of Publication

Enter the year of dataset publication, or completion or completion, of the soil point dataset, in case of unpublished data.

- Average accuracy of point location coordinates (1 to 5)

Enter a class of accuracy among those proposed that reflect the average point location accuracy of the overall project/survey activity. Classes are as follows: 1. GPS coordinates; 2. locations carefully marked on detailed legacy map (e.g. at least 1:25,000); 3. location marked on low detailed legacy map (less than 1:25,000); 4. method unknown, but apparently reliable; 5. method unknown, and apparently not reliable (e.g. place locality or other local features such as elevation, slope, aspect, in sites indicated by the coordinates do not correspond to what recorded on the field sheets).

- Web link

Enter the link to web site of the Project (if any).

- Data Owners

Indicate the names of the persons or entities that own the data (in case of unclear ownership, or of open access data, indicate names of authors).

- Data access restrictions

Indicate if the data have access restrictions. If there are no restriction, write "open access". Otherwise, specify the restrictions that apply.